

## FAQs

### **Q1: Why do simple trading rules often underperform in real markets?**

**A1:** Simple trading rules, such as buying when the price crosses a moving average or selling when the RSI is high, often fail because they are **static and do not adapt to changing market conditions**. Markets are not purely random, and their volatility tends to cluster in certain periods, making fixed rules unreliable over time. For example, a moving average strategy on SPY data underperformed a simple buy-and-hold strategy by a significant margin, highlighting the need for more dynamic modeling approaches.

### **Q2: What is an Autoregressive (AR) model, specifically AR(1)?**

**A2:** An Autoregressive (AR) model is a time series model that **predicts future values based on its own past values**. The simplest form, AR(1), is expressed as:

$$\text{returns\_today} = \alpha + \beta \times \text{returns\_yesterday} + \text{noise}.$$

Here,  $\alpha$  represents the average return without momentum,  $\beta$  quantifies how much yesterday's return influences today's, and noise is the unpredictable part.

- $\beta > 0$  suggests momentum (positive returns follow positive returns).
- $\beta < 0$  suggests mean reversion (positive returns tend to be followed by negative returns).
- $\beta = 0$  indicates a random walk (yesterday's return provides no information about today's).

### **Q3: What is "stationarity", and why is it crucial in time series modeling for trading?**

**A3:** A time series is considered **stationary** if its statistical properties, specifically its **mean (average), variance (volatility), and autocorrelation (relationship with past values) remain constant over time**. In simpler terms, it means "the past looks like the future". Stationarity is crucial because most statistical time series models assume these properties are stable. Suppose markets change significantly (e.g., during crises like the 2008 sub-prime mortgage crisis, the 2020 pandemic, or due to rate hikes). In that case, static models trained on past data will fail if the underlying market structure is non-stationary and constantly shifting. This is why we usually model returns, not prices, as returns tend to be stationary.

### **Q4: How can one test for stationarity?**

**A4:** Stationarity can be checked in two ways:

1. **Visually (Rolling Statistics):** By plotting rolling statistics such as the rolling mean and rolling standard deviation. For a stationary series, these lines should appear relatively flat, indicating consistency over time.

2. **Statistically (Augmented Dickey-Fuller (ADF) Test):** The ADF test determines if a series has a "unit root," which would indicate non-stationarity. The null hypothesis for this test is that **the series is not stationary**. A low p-value (typically less than 0.05) and an ADF statistic lower than the critical values lead to rejecting the null hypothesis, suggesting the series is likely stationary. For example, PayPal stock prices were found to be non-stationary (p-value 0.36635), while PayPal stock returns were stationary (p-value 0.00000).

**Q5: What are ARIMA models, and how do they improve upon simple AR models?**

**A5:** ARIMA stands for **AutoRegressive Integrated Moving Average**. Investment professionals often use them because they address three key problems:

- **AR (AutoRegressive) part:** Captures momentum and mean reversion using past returns.
- **I (Integrated) part:** Handles non-stationary data by applying differencing (e.g., modeling price changes instead of raw prices).
- **MA (Moving Average) part:** Improves predictions by using past forecast errors. This combination allows ARIMA models to be more robust, adapting to changing market trends and learning from prediction mistakes, which simple AR models cannot do as effectively.

**Q6: What is the significance of "rolling forecasts" in designing systematic trading strategies?**

**A6:** Rolling forecasts are a critical component of systematic strategies because **markets constantly evolve, and patterns that worked in the past may fail in the future** (e.g., COVID-19, rate hikes). Instead of relying on a static model trained on old data, a rolling forecast approach continuously retrains the model using a "rolling window" of the most recently available data. This allows the model to adapt dynamically to changing market conditions, addressing the core challenge of "model decay" in real-world trading environments.